Deployers

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1 Deployer overview

An astonishing quantity and variety of CubeSat deployers and configurations have appeared over the past few years. There are in excess of 15 deployer developers that have hardware in space. Nearly every developer has multiple variants of their dispenser (1U, 3U, 6U, 12U, etc.) and varying accommodating features that are available. The accompanying suite of design criteria and attributes per dispenser and dispenser variants can cause a significant amount of uncertainty and prevent progress on the part of the mission. For example, certain deployers can support a 3U z-axis length of 340.5 mm, others can handle 366 mm, and some can support both. Certain deployers can handle the "tuna can" volume; others cannot. Is the protrusion envelope in the x- and y-axes <6.5 mm, <10.0 mm, or some other value? This chapter is intended to act as a transfer function for payload developers to begin thinking through and designing to the dispenser-dependent features and to serve as a pointer to where the latest information not included in this chapter may be obtained.

Boiling things down for simplicity, there are two ways to go about getting a CubeSats manifested to be launched into space. The CubeSat developer can find a dispenser with the features they want and then select a rocket that will get into a desired orbit. This menu approach doesn't completely track, since certain launch vehicles prefer or require certain dispensers or even have contracts with certain launch services companies to handle and coordinate all CubeSat launch opportunities. Alternatively, there are various ride share organizations in which the CubeSat developer can pursue a launch and tailor their design to the dispensers chosen for the ride they would like. Companies such as Spaceflight provide CubeSat to launch vehicle matchmaking and launch services.

1.1 Deployer survey

As is all too frequent in the CubeSat industry, CubeSat developers often need to begin designing and potentially even begin fabrication/procurement prior to knowing which dispenser or launch vehicle they will be using. Very often developers are well into the design and build process without having a formalized vehicle to dispenser/launch vehicle Interface Control Document (ICD). This chapter will attempt to set some general guidelines to allow developers to begin down the path of understanding deployers and the implications to their spacecraft designs. Requisite to a thorough design is an

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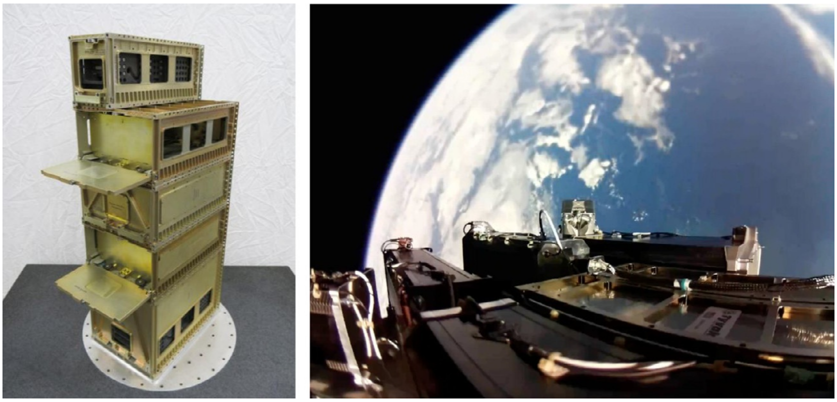

Fig. 1 *(Left)* Planetary systems corporation 3U, 6U, and 12U canisterized satellite deployers. *(Right)* Tyvak 6U NLAS, MK II and 3U RailPOD deployers aboard electron “it’s business time”.
Left: Courtesy Planetary Systems Corporation; Right: Courtesy Rocket Labs.

understanding of what deployers are available and what each offers. Example CubeSat deployers are illustrated in Fig. 1 [1–3].

Getting a payload into space is easier now than ever before. The journey of getting through the proverbial “valley of death” of getting a CubeSat into space is now more of a highway through the desert, due to the infrastructure that has been developed for the CubeSat industry in the form of dedicated launch vehicle and launch service companies. This infrastructure has created a paved pathway of dedicated smallsat and CubeSat launch vehicles, launch service providers, deployer developers, and CubeSat component vendors. The challenge has changed from competing for the very few launch slots on the few deployers to optimizing a CubeSat design to allow for an incredible number of possibilities. The process of solidifying launch opportunities has generated potential revenue opportunities for vendors, spurring the divergence in deployer intracompatibility. Referencing Table 1, it may be seen that there are many deployers from many vendors and permutations of available features (Fig. 2).

1.2 Anatomy of a deployer

A dispenser, or deployer, may be thought of as a jack in the box that acts as a mechanical and electrical interface between a CubeSat and the launch vehicle. Boiling the current state of the art down a bit, once the launch vehicle sends the deployment signal to the dispenser, the door actuation mechanism allows the door to open, which permits the spring(s) to extend the pusher plate toward the now open door, pushing the payload along the rails or tabs and ejecting it from the launch vehicle (Fig. 3).

Table 1 Partial list of available dispensers.

Descriptions	Country of origin	Launch platform	Rails or tabs	Rails/tabs constrained?	Standards
Astrofein/ECM/ EXOLAUNCH PSL-P	Germany	Rocket	Rails	Yes	3U, 3U+, 6U, 12U
Astrofein/ECM/ EXOLAUNCH 16U	Germany	Rocket	Rails	Yes	16U
Cal Poly PPOD	USA	Rocket	Rails	No	3U, 3U+
D-Orbit ION	Italy	Rocket	Rails	No	3U, 3U+, 6U, 6U+, 12U, 12U+
ISIS DUO	Netherlands	Rocket	Rails	No	1U–3U, 6U
ISIS ISIpod	Netherlands	Rocket	Rails	No	1U–3U, Custom
ISIS QuadPack	Netherlands	Rocket	Rails	No	1U–3U,6U, 12U
Jaxa J-SSOD	Japan	ISS	Rails	No	1U–3U
Nanoracks	USA	ISS	Rails	No	1U–6U (1×6)
PSC CSD	USA	Rocket	Rails or tabs	Yes	1U–3U, 6U
Rocket Labs Maxwell	USA	Rocket	Rails	No	3U
SFL XPOD Triple	Canada	Rocket	Rails	No	3U
SFL XPOD DUO	Canada	Rocket	Rails	No	16U
Tyvak RailPOD	USA	Rocket	Rails	No	3U, 3U+, 3UXL
Tyvak NLAS, Mk II	USA	Rocket	Rails	No	3U, 3U+, 6U, 6U+
Tyvak 12U	USA	Rocket	Rails	No	3U, 3U+, 6U, 6U+, 12U, 12U+

Fig. 2 Nanoracks ISS CubeSat deployer.
Courtesy Nanoracks.

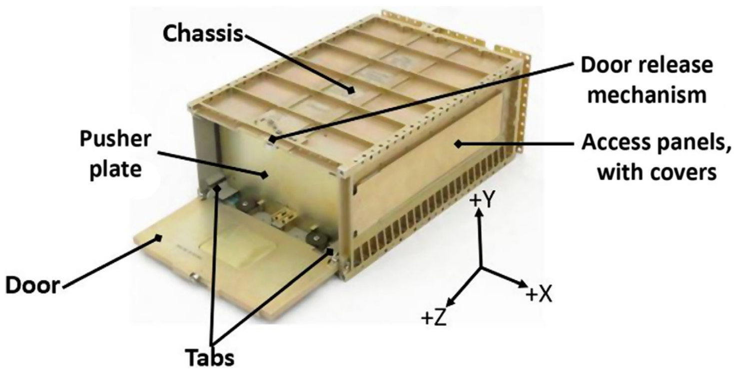

Fig. 3 Planetary Systems Corporation Canisterized Satellite Dispenser.
Courtesy Planetary Systems Corporation.

1.2.1 Deployer door

The door acts as one of the six faces, most commonly the +Z face, that fully contain the CubeSat(s) inside of the dispenser. The door in deployers acts as a constraint that prevents the spacecraft from being ejected into space prematurely. Once the deployment signal is sent to the dispenser the door latch releases, torsion spring forces open the door and allow the CubeSat to be deployed. For most deployers the CubeSat +Z face or rails make physical contact with the door.

1.2.2 Deployer pusher plate

The pusher plate acts as the −Z physical contact location for CubeSats. The pusher plate, as its name implies, pushes the CubeSat along the rails/tabs and out of the deployer once the door has been opened. The pusher plate movement is facilitated by the deployer spring.

1.2.3 Deployer springs

The deployment spring, generally a compression spring, is the ejection mechanism that is actuated when the deployer door opens. The springs extend, forcing the pusher plate and CubeSat toward the deployer door. Generally, deployers utilize a single spring per 3U cavity. The ratio of deployment spring force to mass of the CubeSat determines the deployment velocity.

1.2.4 Deployer chassis

The chassis of a dispenser consists of the five faces that are not the door. One important consideration for CubeSats is to consider the chassis as an isolation mechanism between the CubeSat and the remainder of the launch vehicle. Most dispensers do not permit contact between the CubeSat payload and the chassis. The deployer chassis is the primary interface and load path between the dispenser and the launch vehicle.

Deployer chassis has access panels along the +/−X faces that permit limited access to a CubeSat payload once the vehicle has been integrated into the dispenser, but before the access, covers are integrated.

1.2.5 Deployment mechanism

The deployment mechanism is the device that latches the door to the chassis. It serves to constrain the spacecraft within the deployer and allows it to exit upon command. Once the launch vehicle sends the deploy command, the mechanism actuates and releases the door.

1.2.6 Rails and tabs

The rails/tabs are the primary CubeSat to deployer mechanical interface. These features constrain the CubeSat in the X-Y plane, and certain dispensers constrain Z-axis movement as well. There are two variants: rails and tabs. The rail system utilizes the four corners of the CubeSat for constraint. The tab systems utilize the +X, -Y and −X, -Y corners to constrain the CubeSat. The deployer tab interface clamps down on the CubeSat tabs to provide an invariant load path and constrain motion in the X, Y, and Z axes.

2 Launch services requirements and documentation

Launch services have become a viable industry and serve as the intermediary between the payload developer, deployer developer, and launch vehicle developer. There are several companies that offer these services.

There is an adage for developing spacecraft that indicates that required documentation stacked up is taller than the satellite itself. While the data products required for CubeSat missions are generally much less than for larger spacecraft, there is still a considerable volume. The documents required to be provided to the launch service contractor vary considerably and are dependent upon the launch vehicle being utilized. The following list outlines the most common data products required to be provided by the developers:

- mass properties statement
- venting document
- materials list
- orbital debris assessment review (ODAR)
- transmitter survey
- frequency coordination documentation
- thermal vacuum test report
- dynamics (random vibration, sine sweep, sine load vibe, shock, etc.) test report
- CubeSat electrical requirements report
- CubeSat mechanical requirements report
- CubeSat compliance memo
- day-in-the-life test report

3 Deployer characteristics

The CubeSat form factor, now 20+ years from its conception, has grown and formed an entire industry. This industry, as it relates to dispensers, has continued to innovate once the powers that be have become accustomed to the idea of a CubeSat. This innovation has created a variety of different designs from both the private and government sectors. To shed some light on what the standards and features are, it should be noted that deployer developers have followed the spirit, if not the letter of the "CubeSat standard" [1,4].

3.1 Deployer to CubeSat energy translation

In the process of getting a CubeSat to space, the integrated CubeSat dispenser system is mounted to a rocket. The process of getting to space and the space environment are not benign. A payload will be shaken and baked during the ascent before being dispensed from the launch vehicle. The launch services provider will provide the anticipated dynamic and thermal environments the dispenser should expect to see. Note that this is not what the CubeSat will experience, but the conditions that the dispenser will experience, most often defined at the launch vehicle to dispenser interface. Every rocket and launch profile is different and will expose the CubeSat to a different environment. The question the CubeSat developer should ask is "Now that we are manifested and have anticipated loads for the deployer, what will the satellite experience? Do the loads published in the ICD have any margin included? If so, how much?" The satellite developer should consider both the thermal and dynamic energy translation through the dispenser to verify that the CubeSat design is adequate.

3.1.1 Thermal translation

When anticipated thermal values are indicated for a mission in a mission-specific ICD, they are generally indicated for the temperatures that the dispenser surface will experience with some amount of margin. Note that these are not the temperatures that the CubeSat will experience. One can think of the thermal mass and the thermal interface between the rocket and the dispenser as a filter or a thermal buffer. The thermal transportation processes in space, once above the Karman line, are limited to conduction and radiation; additionally, the CubeSat should be isolated from short-term temperature radiative effects due to the fact that they are completely encapsulated. If the launch vehicle is intended to BBQ roll, then temperatures will be further muted from the extremes.

The temperatures experienced pre- and short-term postdeployment can vary wildly. Think through the impact of deploying hot or cold, since CubeSats will not likely have an option as to where the rocket deploys the payloads. Generally, in LEO, this is determined by whether the deployment will occur while the dispenser is illuminated by the Sun or eclipsed by the Earth. Deployment mechanisms tend not to function as well if the deployment is actuated while the mechanism is cold. This will depend upon how well the materials comprising the mechanism coefficient of thermal expansion (CTE) have been matched. Testing deployments at worst-case cold temperatures, building in sufficient deployment margin in movement mechanisms, or including temperature measurements of strategically placed thermal measurement devices in when the mechanism is deployed can buy down the risk of non- or partial deployments.

3.1.2 Dynamic energy translation

Similar to the thermal load specification, random and static loads are most often specified for the dispenser. This information is not a good indicator of what the CubeSat payload will experience. The deployer will act as an energy translator between the launch vehicle (LV) and payload. Remember that the deployer will have resonant modes as well. Also of note is that many developers have begun to include internal and/or external vibration dampeners to reduce loads applied to the payload.

Through many vibration tests the author has found that railed dispensers tend to dampen sub- 100 Hz energy, amplify low-hundred Hz energy, and attenuate in the mid-hundreds to high hundreds of Hertz through low kiloHertz ranges. Keep in mind that this does not take into consideration dispenser or vehicle resonances. Also the slight gap between the CubeSat and dispenser rails permits the rail-based CubeSats to bounce around while exposed to dynamic loads. Tab-based and some rail-based dispensers clamp onto CubeSats, preloading/stiffening the payload to dispenser interface and preventing movement. This causes clamping rail systems to increase resonant mode frequencies, when compared with nonclamping systems.

3.2 CubeSat deployment tip-off rates

Once the door is opened and the CubeSat is ejected from the deployer, the payload will not come out straight and stabilized. This is due to a number of factors including spring force; LV roll rate, whether tabs or rails are used; CubeSat orientation inside of the dispenser; mass properties; and other considerations. The rule of thumb for most dispensers is that CubeSats should expect a one-sigma rotation rate of 10 degrees per second per axis. There have been several studies conducted of rotation rates in representative environments such as in [5]. This study outlines empirical rotational rates of a payload ejected from a 6U CSD.

The CubeSat developer should consider the first boot order of operations and gauge the impact of imparted dynamics including tumbling and rotation. For example, does the spacecraft have large deployables? If so the developer may want to consider whether these deployables can withstand the given rotational rates. Perhaps the rotational rates could be dampened prior to initializing deployment. If the rotational rates are not well characterized, and the spacecraft contains an attitude determination and control system (ADCS), developers should attempt to get an attitude solution and determine the rotational rates. If they are excessive, attempts should be made to dampen them prior to solar array, antenna, etc. deployments.

3.3 Rail and tab considerations

The rails or tabs are meant to be the sole mechanical interface between the dispenser and the CubeSat; accordingly, these areas should be considered keep-out zones. There is generally a deployer-specific exclusion area around these interfaces. The CubeSat developer should consider whether the worst-case bias inside the deployer will interfere with these contact zones.

Rails and tabs are generally required to be either anodized Al 6061-T6 or Al 7075-T7. There have been other flight-verified materials for various deployers such as certain additively manufactured materials such as Windform XT2.0 [6]. Should the developer desire to utilize a previously unqualified material, they should expect to undergo a material qualification regimen that includes the following:

- Verification of the strength of the material
- Verification of CTE and performance analyses of impact to dispenser material
- Verification of material surface roughness
- Verification of size differential pre-/postvacuum testing
- Verification of material outgassing
- Verification that the material is nonconductive

Rails and tabs of both the CubeSat and dispenser are required to be anodized. Historically, this was meant to prevent cold welding between the CubeSat and dispenser. This anodization also results in electrical isolation between the payload and deployer. These areas should be considered critical dimensions by developers. Accordingly, developers should make sure to factor in the dimensional change from anodization after fabrication in the tolerance stackup. Rails for 1U through 3Us are generally required between 100.0 and +/− 0.1 mm in the X and Y planes. Should the structure be designed to 100.0 mm exactly, this requirement could quite easily exceed the requirement, especially when conforming to MIL-A-8625 [7], that generally results in a 0.025–0.05 mm of buildup per surface. Anodizing these areas will cause buildup on each rail, causing twice the effect for any given axis measurement. The buildup for tabbed systems should similarly be verified. The tabbed systems generally are required to be between 2.95 and 3.05 mm, so designing the tabs to be exactly 3.00 mm would almost certainly exceed the requirement, depending upon the machining tolerances.

As the deployer door is opened, the CubeSat will slide along the dispenser rails/tabs. Should the interface be insufficiently smooth, excessive friction between surfaces may cause nondesired effects, such as greater than normal rotational rates, slow, or even failed deployments. Rail and tab average surface roughness (Ra) is most often specified as $\leq 1.2\,\mu m$ postanodization [8]. Another related consideration, generally for nonconstrained systems, is that when CubeSats are dynamically tested (random vibe, sine load vibe, etc.), developers should expect some wear on the rails. This wear creates small amounts both flaking of the anodization and metal shavings/powder and is most prolific for the rail corners and Z faces. Beyond the hazards of this FOD, the rails may be deformed. Careful sizing and tolerances of the rails may reduce the chance of this occurring.

4 CubeSat to deployer testing

CubeSat payloads are expected to undergo a testing regimen that is somewhat variable based upon the launch vehicle. The specific tests outlined in Fig. 4 are common across virtually all missions as they pertain to deployers [7]. Assuming no waivers are required against the CubeSat to dispenser ICD, few other tests are required, again dependent upon mission, to confirm compatibility with the dispenser to be used for the mission.

4.1 CubeSat to deployer fit check and CubeSat acceptance checklist

The tests that fall under fit check and CubeSat Acceptance Checklist (CAC) cover mechanical measurements such as payload envelope, mass, switch and separation location and functionality, rail size and keep outs, and protrusion envelope. This is something that should be measured early and often with calibrated measurement devices if possible. If calibrated tools are unobtainable, developers should attempt to make the same measurements with multiple sets of measurement devices. The launch vehicle integrator will likely measure the payload prior to integration. If the tools utilized indicate that the payload does not conform, the CubeSat will be rejected. Strictly speaking, any changes to the vehicle at this point invalidate all CubeSat-level tests; therefore the payload may be required to reperform vibration and thermal vacuum tests. The mission-specific ICD will contain the dispenser/launch vehicle restrictions. As is often the case in the CubeSat world, an ICD is not in place until late in the vehicle development flow. If this is the case, developers need to make sure to obtain the dispenser-specific requirements. Alternatively, use the CACs that may be obtained at CubeSat.org until mission-specific details are provided.

Generally the launch service provider will provide a model to developers that will allow them to perform a vehicle to dispenser fit check. If possible, the team should schedule this as early as possible. If no model is provided, developers should attempt to make a simple alternative model, or be very certain of the measurements that indicate compliance. Arriving for integration activities and discover that the spacecraft does not fit will jeopardize the mission's certification of flight readiness.

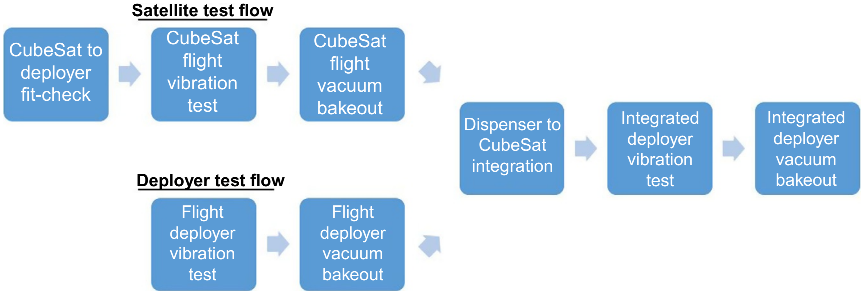

Fig. 4 General CubeSat to deployer test regimen.

4.2 Dynamics testing

Dynamics testing is a rather large umbrella that incorporates a number of tests. It includes everything from low-frequency sine load tests, sine burst, characterization sine sweeps, random vibration, to shock testing. All launch vehicles require random vibration testing and sine sweeps. The loads that the dispenser are anticipated to experience during flight will determine whether a requirement will be levied for the other tests. Tests such as acoustic testing and centrifuge or sine burst static load tests are generally not levied due to the fact that CubeSats are so small.

Most missions will require a standalone CubeSat vibration test of the flight unit, prior to integration into the dispenser. The payload will be required to test the unit to the levels indicated in the mission-specific dispenser to CubeSat ICD. Remember, as is indicated in Section 3.1.2, the levels define the levels for the dispenser to launch vehicle interface and have no bearing upon the payload. The payload may experience more or less energy than the levels experienced by the dispenser, depending upon the design of the payload. CubeSat developers should expect a second and generally lower-level test to be performed once the CubeSat has been integrated into the dispenser.

Test fixtures for dynamics testing are very important. Generally the launch service provider or deployer dispenser will have a dynamic testing system, most often called a TestPOD. In the event that no test fixture is provided, it will be the responsibility of the payload developer to design and fabricate something or obtain a similarly designed test design from another vendor. It is important to keep in mind that all deployers are not created equal. A TestPOD from a different dispenser vendor will likely create a different ride/environment/experience for the payload and is therefore not ideal.

4.3 Thermal vacuum testing

Thermal vacuum tests are most often a nonpowered thermal vacuum bakeout process. In other words, they are a process of to baking off volatile contaminants that are present from the vehicle assembly and testing processes. The most frequent levels are defined as maintaining a vacuum $<1 \times 10^{-4}$ Torr and profiles of 60°C for 6h, 70°C for 4h, or 50°C for 24h [7]. Most developers tend to perform this process at the beginning of other thermal vacuum testing such as thermal cycling and thermal balance. Although somewhat rare, additional thermal vacuum tests are levied on the CubeSat by the dispenser ICD. Fig. 5 shows common thermal vacuum bakeout temperature profiles.

5 Additional CubeSat design considerations

Satellites are designed to fly in space; however, if developers do not consider the deployer requirements and focus only on those associated with the satellite, the design might not be viable for a specific deployer/launch vehicle or effectively achieve orbit. Experienced satellite developers know that they need to design certain features into their spacecraft such that it may be tested and assembled. Similarly, CubeSat designs

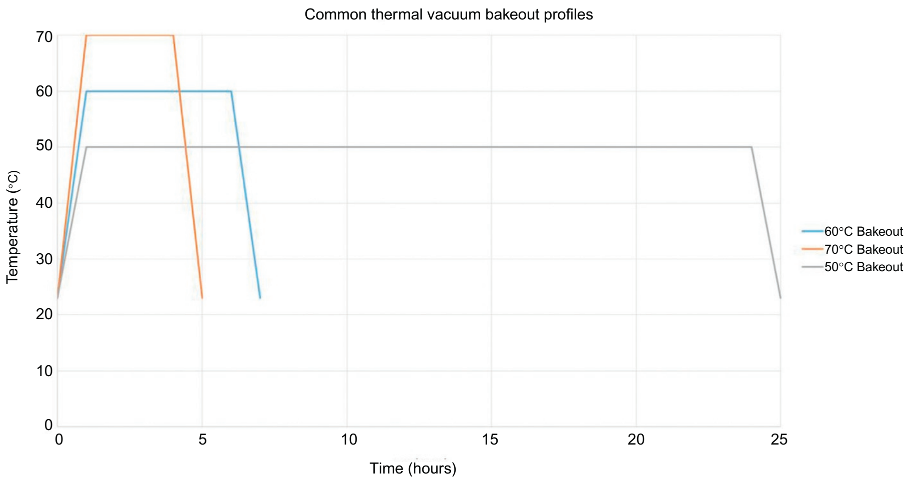

Fig. 5 Common thermal vacuum bakeout temperature profiles.

should be tailored to the dispenser in which they intend to fly, considering the following elements:

- center of mass/mass properties
- mechanical envelope
- electrical interfaces
- separation springs
- inhibit switches
- remove/insert before flight article locations
- CubeSat to dispenser integration
- CubeSat postintegration and prelaunch delays
- CubeSat venting
- CubeSat initial contact and LEOP concept of operations

5.1 Center of mass

Beyond keeping track of the center of mass (CM) for attitude determination and control purposes, the launch service team will need to know where it is located. The premise for launching CubeSats for most launch vehicles is the concept of replacing ballast mass with payload mass and generating additional revenue. The ballast is placed in a manner that corrects the launch vehicle center of mass; therefore the launch vehicle will need to know where the combined CubeSat/dispenser center of mass is located. Since the payload mass is generally equal to the mass of the dispenser, the CubeSat CM factors quite heavily into where the *system* (CubeSat + Dispenser) center of mass is located. The center of mass requirement varies per dispenser, but the boiler plate requirement is between 6.5 [1] and 10 mm [4] of the CubeSat geometric center. Modern-day dispenser requirements can accommodate center of masses of ≥ +/− 20 mm [1,4].

It is incredibly difficult to correct the center of mass with the limited mass and volume inherent to CubeSats. While designing the spacecraft, developers should keep an eye on where the center of mass is located and reserve certain areas to correct X, Y, and Z centers of mass. Developers should attempt to identify which dispenser is being utilized as early in the design process as possible, and utilize the indicated center of mass reference point. Common points are the −X, -Y, and -Z corner and X-Center, -Y, -Z point. The rationale for the importance is that tracking multiple reference points for attitude determination and control becomes confusing and can become labor intensive to maintain multiple sets of mass properties. It is often a good idea for those maintaining the CAD model to make a box or sphere with zero mass that encompasses the allowable center of mass.

5.2 Mechanical envelope

It is easy to lose track of the mechanical envelope requirements in the midst of designing a CubeSat. There are quite a few to keep in mind. It can be useful to create a max envelope within a CAD model that defines all mechanical envelope requirements. It should include details such as rail or tab maximum allowable dimensions; protrusion envelopes; inhibit switch required locations; separation switch locations; appropriate contact area with the door, pusher plate, and rails/tabs; and access panel locations. Using this in combination with the deployer CAD, if accessible, can prevent designing a satellite that will not conform to requirements. Additionally, developers should keep in mind when using unconstrained deployers that the protrusion envelope needs to be able to accommodate worst-case biasing of the CubeSat in the x and y-axes. In this case, the CubeSat will bounce around inside of the dispenser during ascent and in-orbit maneuvers of the launch vehicle. This effect should be accounted for in the CubeSat design.

5.3 Electrical interfaces

The electrical interface design considerations consist of several facets. The design of many dispensers allows direct access to the CubeSat GSE charge and data ports, remove before flight (RBF), and/or insert before flight (IBF) while the CubeSat is integrated. Most deployer designs do not allow for a direct electrical connection between the CubeSat and dispenser and those that do require precise dispenser to CubeSat connector alignment.

Should the CubeSat developer desire the ability to test or verify the vehicle status once integrated into the dispenser or between axes during dynamic testing, the developer should make sure to place the charge and data ports such that align with the access panels. While performing final integration of the CubeSat into the flight dispenser, designers need to make sure to think through the process to prevent the vehicle from powering ON. One best practice is to design an RBF or IBF article that will allow for integration into the dispenser while these articles are in place, and once integration is complete, the articles are accessible via one or more of the access panels. Also of note is the very real possibility of launch delays, which means that several months or even years could pass before launch. In situations such as this, it is common for the CubeSat to be allowed to recharge if requested after 3 or 6 months. If the charge port is not within the access port area, then no charging is possible.

5.4 Separation springs

Separation springs are fairly straight forward. Should there be multiple CubeSats within a given deployment volume, separation spring are used to create separation between the various payloads. Should there be multiple payloads within the same dispenser, one low-cost method of achieving semicontrolled separation is to use calibrated springs to achieve the relative velocity between the payloads. Also of note is the ability to overcome other payloads within the same 3U volume that either prematurely deploy before deployment or have excessively strong magnets. A sufficiently strong separation spring will prevent payloads from being struck together. Most launch service providers will analyze all permanent magnet strengths at the CubeSat external surfaces to verify that will not be stuck together upon deployment; however, CubeSats have been deployed while inadvertently coupled together in flight before. The industry standard separation spring strength is 0.9lbs (4 N) [1].

5.5 Inhibit switches

Inhibit switches are mechanical switches that ensure the vehicle remains powered off while one or more are mechanically closed and while integrated into the dispenser. There are generally several requirements as to their number and location. The switches are generally the only CubeSat articles, other than the rails/tabs and separation springs, that may contact the dispenser or other payloads within the same cavity. Inhibit switches are most often required to be located on the −Z side of the vehicle and make contact with the pusher plate. Some missions allow roller switches to ride along certain portions of the X or Y faces of the deployer. Generally, two or three independent switches are required.

As has been mentioned previously, the CubeSat moves around within unconstrained deployer systems during vibration tests and during ascent. This and deployment spring movement can allow the inhibit switches to chatter or open and close quickly. CubeSat developers should design their power and computer systems to both allow for this by creating a small time lag between switch actuation and boot and by making certain that any deployment timers to reset in the event of a power on event.

5.6 Vehicle to dispenser integration

CubeSat developers should think through how their vehicle will be integrated into the dispenser. Should the CubeSat be integrated with the dispenser oriented vertically or horizontally with reference to the orientation of the door? Designers should recall that unless the dispenser includes a spring locking mechanism, the integrator will be fighting the spring during integration. Integrating horizontally allows for easier initial integration but adds more risk to the payload in the event the integrator loses a handle while integrating. This could potentially push the payloads back out and potentially causing damage. Do the RBF articles have sufficiently low profile to remain within the protrusion envelope, or will they come into contact with the dispenser? If so, is there a way that will allow for a partial integration, ensure deployment switches are actuated before removing the RBF pin, and for the remainder of integration activities? Is there GSE to verify alignment with the dispenser cavity? The dispenser cavity is fairly tightly toleranced and it can be difficult to prevent damage while inserting the CubeSat.

5.7 CubeSat postintegration, prelaunch delays

Launch delays are very common. Once payloads are integrated into dispensers, developers generally no longer have access to their CubeSat. In the event of a post-integration delay, the CubeSat power supply should have sufficiently low parasitic current consumption that would allow for 6 months of delay and still be able to recover once deployed in space. The industry standard is to allow for spacecraft charging after this period of time; however, it is not guaranteed. This charging opportunity will only be available for those who have placed their charging port in the access panel designated locations. Another alternative is to design the power system such that if the CubeSat were to be launched with dead, it could recover by charging as the CubeSat tumbles, postdeployment. Similarly, stowed deployable mechanisms should be able to be stowed for long periods of time without permanent deformation or compromising the mechanism.

5.8 CubeSat venting

Venting is rarely a problem due to the size of a CubeSat; but it is something developers will need to address. While a payload is in transit between the launch pad and space, all nonhermetically sealed containers will experience a pressure change of 760 Torr at the launch pad to something along the lines of 10^{-10} Torr in the period of a couple minutes, with the majority of it occurring in a few seconds [7]. If venting is not taken into consideration, pressure concentrations can occur in nonload-bearing structures, causing damage. The industry standards are to design the system to either (1) have an adequate ratio of empty internal volume and pathway to surface area for air to escape, or (2) to have a venting rate above some gas flow rate threshold in pressure versus time ($\Delta P/\Delta t$). The open surface area to volume requirement often used is less than 2000-inch requirement, that is the ratio of empty internal volume in inches^3 to direct pathway surface area for the air to escape in inches^2 should be <2000 inches. Should the requirement be in metric units, the requirement is less than 5080 cm. The volumetric rate calculation can be found in Chapter 13 of reference [2].

6 Conclusion

The innovation permitted by the advent of the CubeSat industry ensures an ever-changing list of dispensers that are commercially available with an ever-increasing list of features and accommodations. There are many dispensers not included in this chapter due to the requirement to limit the content. Also not included is the mapping of standards in which each dispenser can support. Some dispensers are capable of supporting a 365.9-mm Z-axis, while others are capable of supporting the more traditional 340.5 mm. Some dispensers are capable of supporting a 10.0-mm protrusion envelope and others 6.5 mm. In an attempt to keep this text relevant to the constant changing state of the art, this content has been intentionally not included. Instead, information has been provided to assist that developers, regardless of experience, will have the background and references needed to pursue the most up-to-date information for the various dispensers and their permutations.

References

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